CSE 309 Theory of Automata, Spring 2026
Exercise 4

1. **Interpreting Context Free Grammars** Answer each of the following questions for the CFG G given below.

$$
\begin{aligned}
R &\to XRX \mid S \\
S &\to aTb \mid bTa \\
T &\to XTX \mid X \mid \lambda \\
X &\to a \mid b
\end{aligned}
$$

   (a) What are the variables and terminals of G? What is the start symbol?
   (b) Given three examples of strings in $L(G)$.
   (c) Given three examples of strings *not* in $L(G)$.
   (d) Is it true that $T \Rightarrow aba$?
   (e) Is it true that $T \Rightarrow^* aba$?
   (f) Is it true that $XXX \Rightarrow^* aba$?
   (g) Is it true that $S \Rightarrow^* \lambda$?
   (h) Give a description in English of $L(G)$.

2. **Derivation and Parse Trees** Consider a CFG with variables $\{E, T, F\}$, terminals $\{+, \times, (,), a\}$, start symbol E, and the following rules.

$$
\begin{aligned}
E &\to E + T \mid T \\
T &\to T \times F \mid F \\
F &\to (E) \mid a
\end{aligned}
$$

   Give a parse tree and a derivation for each of the following strings.

   (a) a
   (b) $a + a$
   (c) $a \times (a + a)$

3. **Design of CFGs** Give CFGs that generate the following languages with the alphabet $\Sigma = \{0, 1\}$.

   (a) $\{w \in \Sigma^* \mid w \text{ contains at least three 1's}\}$
   (b) $\{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}$
   (c) $\{w \in \Sigma^* \mid \text{length of } w \text{ is odd}\}$
   (d) $\{w \in \Sigma^* \mid \text{length of } w \text{ is odd and its middle symbol is } 0\}$
   (e) $\{w \in \Sigma^* \mid w \text{ contains more 1's than 0's}\}$
   (f) $\{0^n 1^m 0^n \mid n, m \geq 0\}$
   (g) The empty set.